

Project Name: A.C.E (AI Casino Education)

Team Members: Colby Fournier, Cisco Harbeck, Reymond Sanchez

Projected Overview:

A.C.E is an AI-assisted Blackjack training game that helps users learn Blackjack strategy, probability, and decision-making in a low-risk interactive environment. Players will engage in fully playable games of Blackjack while receiving real-time AI-driven strategies that change depending on the user's current hand and the dealer's cards. The core educational feature is the AI-driven strategy engine, which suggests the most statistically optimal move for every possible hand combination. After each move, if a less than optimal one is made by the user, the AI will explain why the move was not optimal, which promotes learning through feedback. Player statistics, such as a win/loss record, and session history will be stored in a local database, which can then be accessed by users to track their progress. If time allows, additional casino games will be implemented, however Blackjack will remain as the primary deliverable.

Expected Software/Hardware to be used:

Software:

- Git/GitHub
- Visual Studio Code
- Python
- SQL/SQLight

Hardware:

- Personal computers/laptops

How is it related to the networking/computing field?

Our project relates to the computing field because it involves software development, algorithms, probability calculations, AI-assisted decision making, backend logic, and database management. It also uses concepts commonly found in computer science and software engineering.

Potential impact/market:

Our project could be used as a learning tool for users who want to improve their blackjack strategy (or strategy for other potential games) and better understand probability / statistics. This project can also demonstrate how AI-assisted systems can analyze information and make recommendations based on data.

Anticipated roles:

Colby: Game logic, UI design and layout

Cisco: Database integration, back-end, testing, and assisting with game systems/logic

Reymond: AI-system design